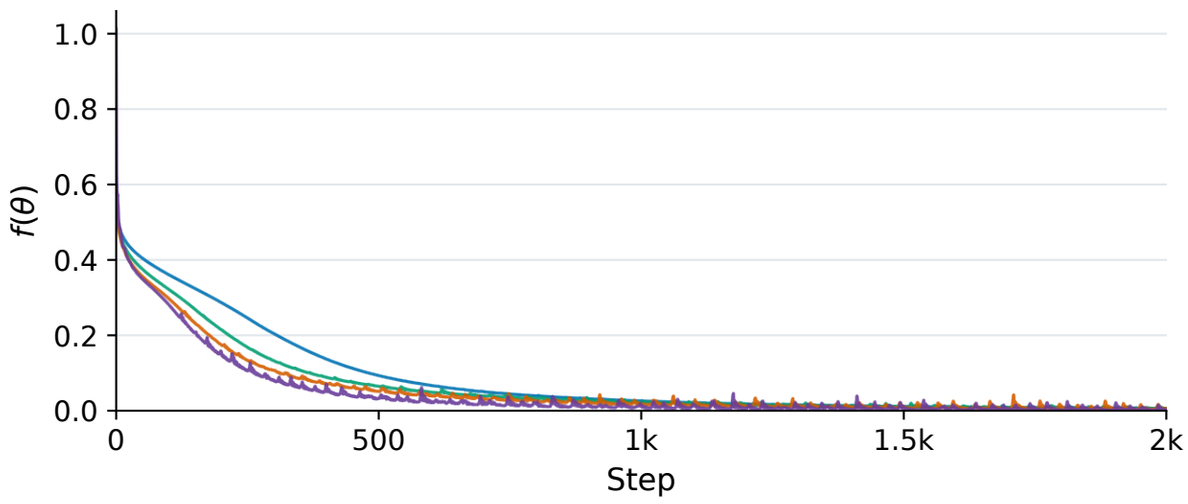
CIFAR-10 | Full-batch Adam

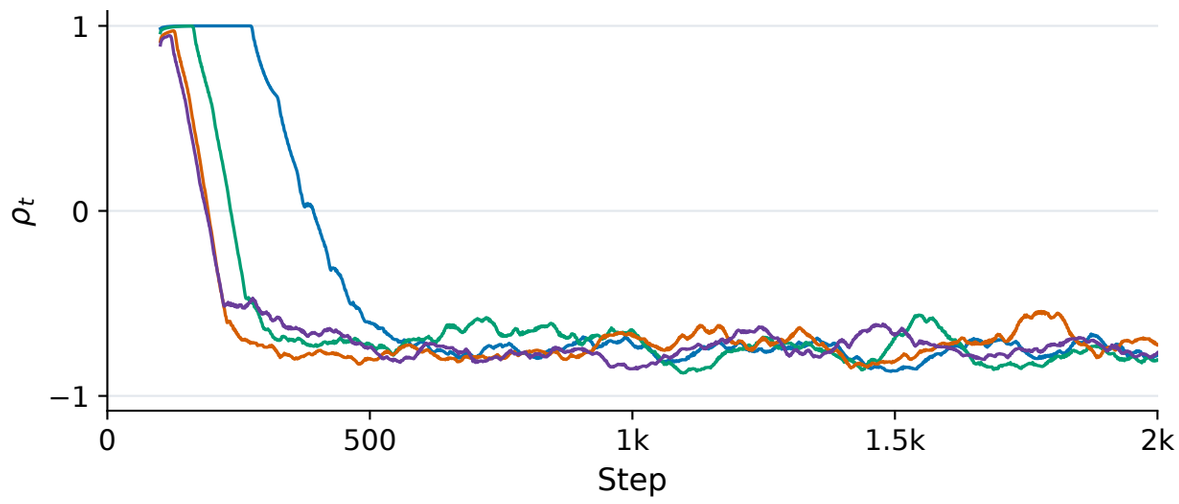
5,000 training images | tanh 200-200 MLP | First 2,000 updates

$\eta = 5 \times 10^{-5}$ $\eta = 10^{-4}$ $\eta = 2 \times 10^{-4}$ $\eta = 4 \times 10^{-4}$

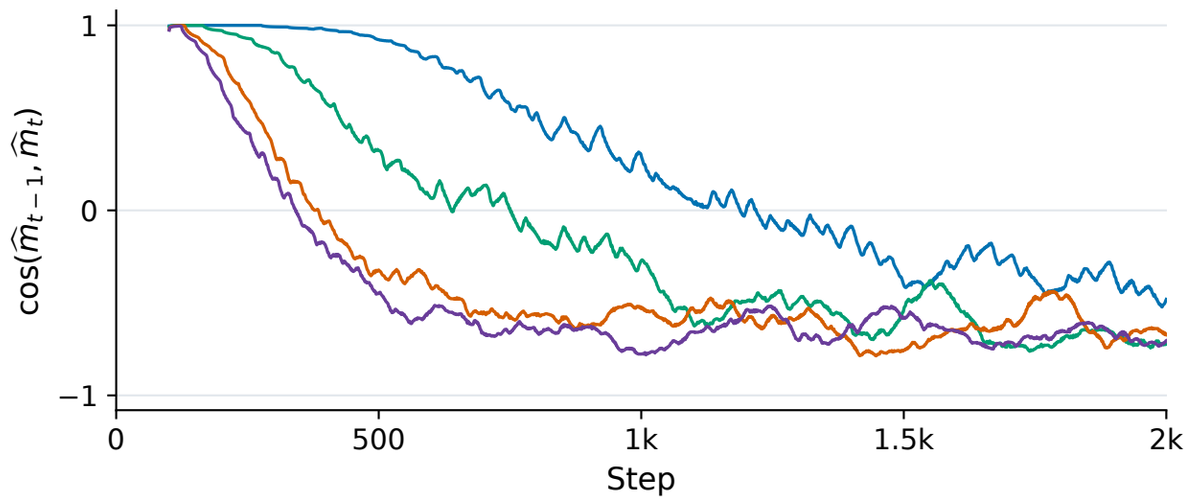
(a) Unsmoothed training loss



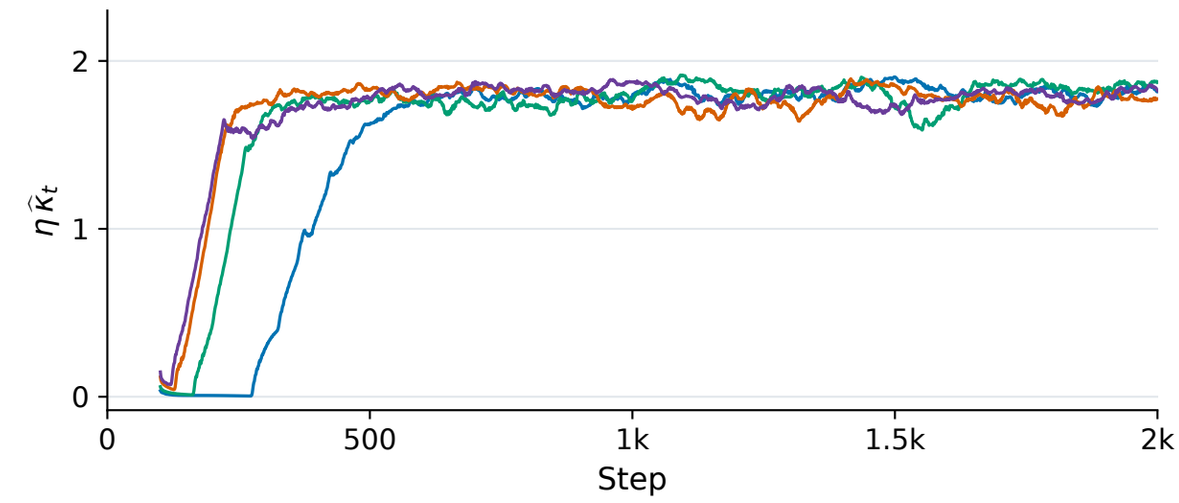
(b) Raw-gradient adjacent cosine



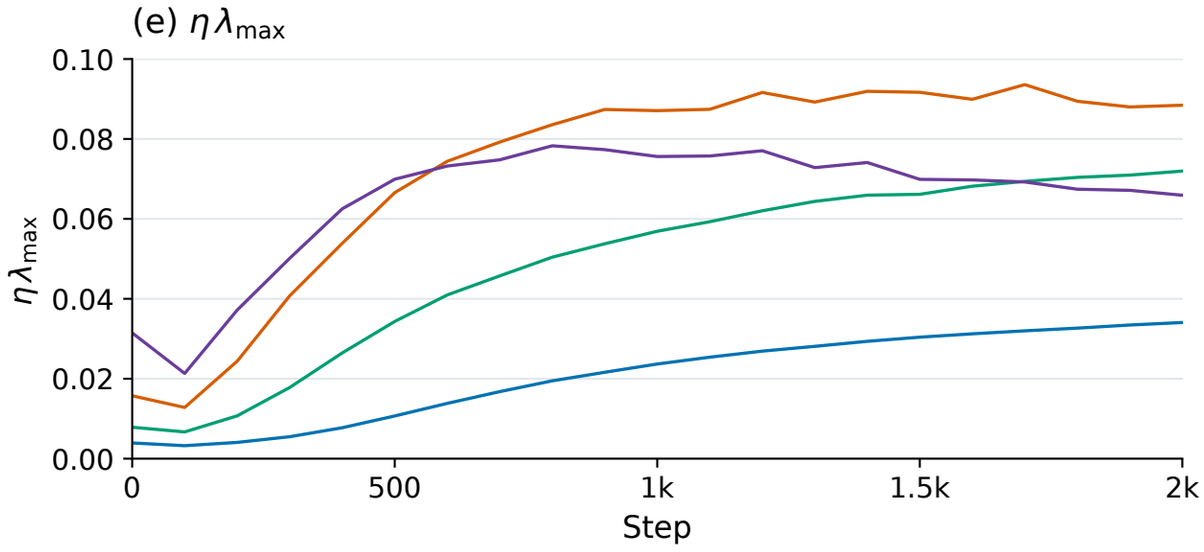
(c) Momentum adjacent cosine



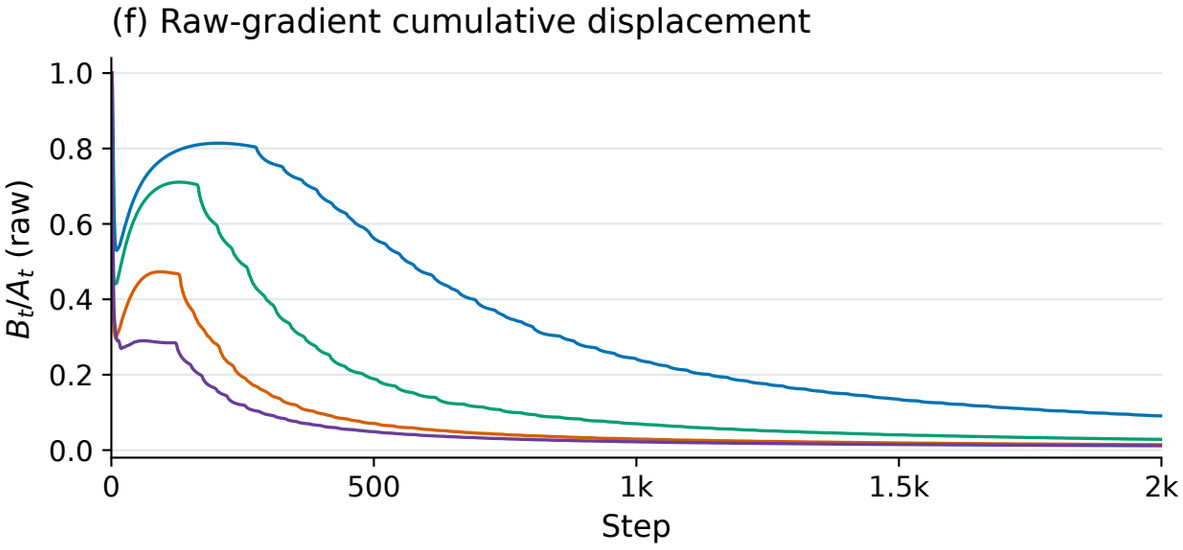
(d) $\eta\,\widehat{\kappa}_t$



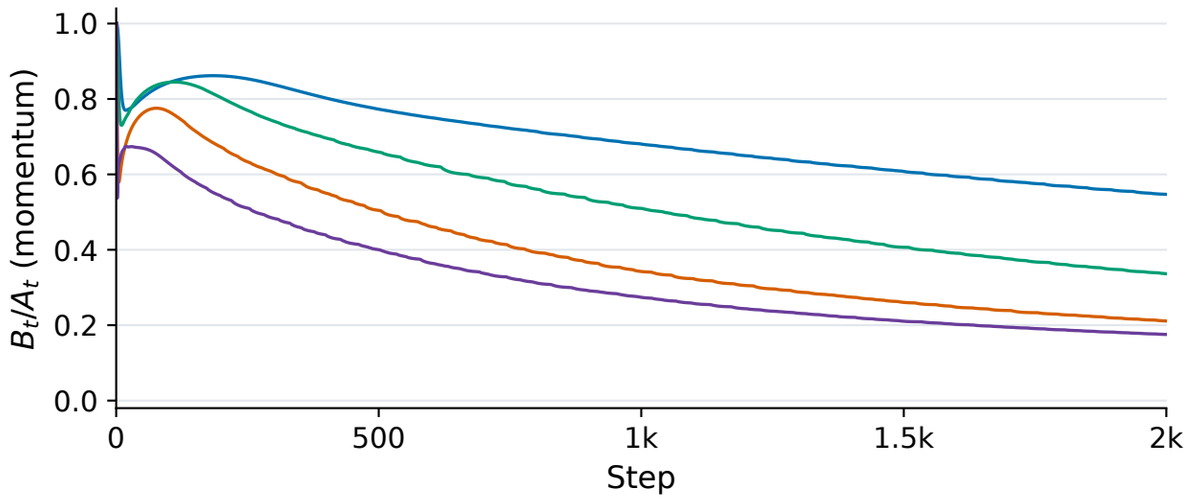
(e) $\eta\,\lambda_{\max}$



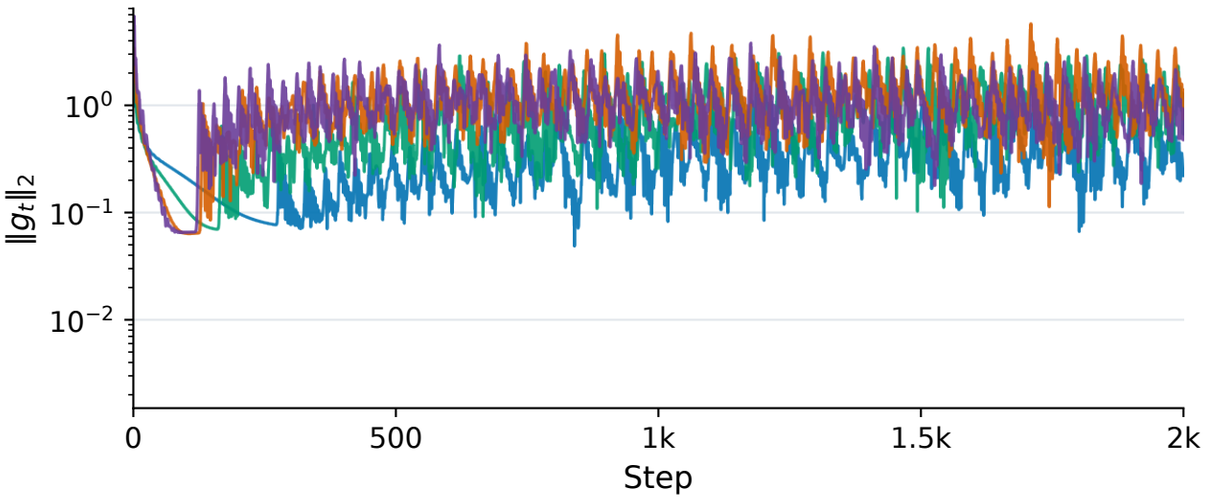
(f) Raw-gradient cumulative displacement



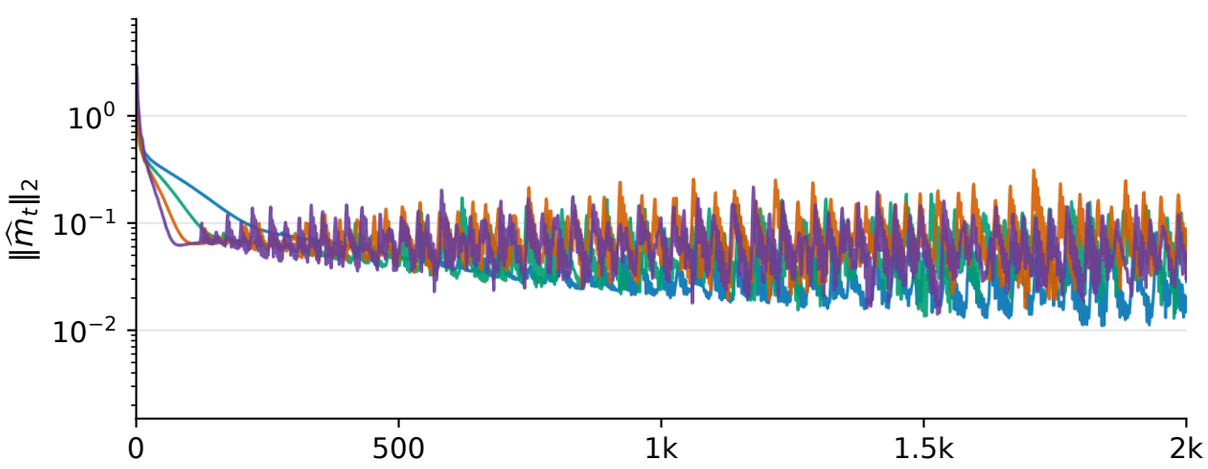
(g) Momentum cumulative displacement



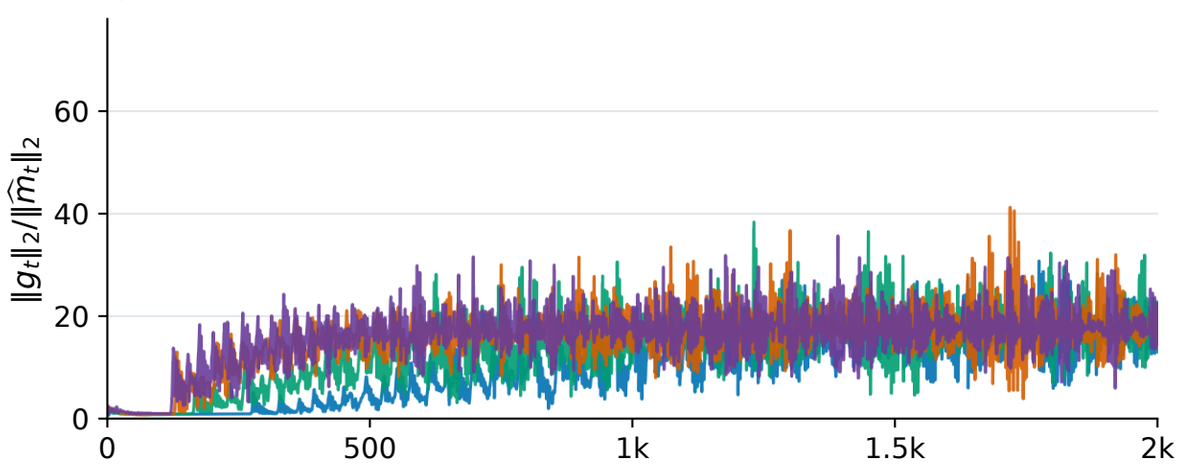
(h) Raw-gradient norm



(i) Momentum norm



(j) Gradient / momentum norm



Momentum is bias-corrected: $\widehat{m}_t = m_t/(1-0.9^t)$. Cosines and $\eta\widehat{\kappa}$: 100-pair trailing means.

Loss, norms and norm ratio are unsmoothed. Only norm panels use log Y; full and early views share Y scales.

Ordinary Hessian every 100 steps. $\eta\widehat{\kappa}$ is a gradient trajectory proxy along Adam.